ROUND 2 BUILD SUBMISSION | WORKING MVP DELIVERED | TRACK: PROBLEM STATEMENT 01 | TEAM INNOVIX

RelayState: Round 2 Project Documentation

A Grounded Knowledge-Handover Engine & Cryptographic Audit Ledger

GITHUB REPOSITORY	DEPLOYMENT MODE	TEAM LEAD & ENGINEER	TEST SUITE STATUS
github.com/kavin-77/relaystate	FastAPI + SQLite + Docker	Kavin S S G (Solo Builder)	All 6 Pytest Suites Passing

01 Working Core Workflow & User Journey

30 Points

In Round 2, the complete 5-stage product journey operates end-to-end. Evaluators can launch the application, load the synthetic corpus (*Project Hyperion*), inspect grounded claims, and complete the transition from leaver drafting to successor continuation:

STAGE	FUNCTIONAL CAPABILITY	EVALUATOR INTERACTIVE JOURNEY
01 Connect	Ingests 56 commits, 25 PRs, 121 chat messages, and 13 Jira tickets.	Click `Load Demo Project`. Ingestion runs in <1 second with real-time stream status indicators.
02 Reconstruct	Extracts decisions, blockers, dependencies, and bus-factor concentration.	Automatically surfaces that Dave authored 97.7% of `billing/` code as a critical organizational risk.
03 Verify	Verbatim grounding check + temporal cross-modal supersession check.	Chat Redis agreement is flagged `SUPERSEDED_BY_CODE` by Memcached commit; ungrounded quotes are dropped.
04 Review	Split-screen review studio with custom redaction tool and leaver sign-off.	Leaver edits claims, redacts strings across trails, and executes digital sign-off (promotes `v0.1 → v0.2`).
05 Handover	Lead signs off `v1.0`, unlocking successor Day-1 view.	Successor receives `HTTP 403 Forbidden` until Lead signs off; once published, successor acknowledges receipt.

02 Changes from Round 1 Plan & Justifications

20 Points Progress

COMPONENT	ROUND 1 PLAN	ROUND 2 IMPLEMENTATION	ENGINEERING JUSTIFICATION
Architecture & Stack	Next.js 14 + Postgres + ChromaDB	**FastAPI + SQLite + Vanilla JS (Single HTML) + Docker**	**Self-Contained Deployment:** Eliminates external DB connection pool drops and client hydration issues for evaluators. Runs as a single process that boots in under 1 second.
Authentication	Mandatory OAuth 2.0 Login	**No-Login Role Picker + Dismissible Info Banner**	**Accessible Evaluation:** Evaluators can inspect all 3 personas immediately without account creation friction. Access boundaries are enforced by the server based on the active role header.
Data Isolation	Shared session cookies	**Per-Visitor Workspace (UUID + `X-Workspace` header)**	Allows multiple evaluators to test simultaneously in isolated sandboxes with a dedicated "Reset" button.
Audit Ledger	Abstract relational audit log	**Cryptographic Hash Chain with Tamper Simulation**	Interactive proof: Evaluators can click `Simulate Tamper` and observe verification detect tampering, then click `Restore`.

03 Technical Depth, Security & Planted Evaluation Traps

25 Points

Planted Evaluation Traps:

- **Temporal Supersession Trap:** Oct 10 chat agreement to use Redis for session cache is compared chronologically against Git master code. Oct 18 commit (`commit-7a89b1`) implementing Memcached automatically demotes the chat claim to `SUPERSEDED_BY_CODE` .
- **Pre-Storage Secret Scrubbing:** Ingestion runs regex/NER token filters that sanitize AWS access keys (`AKIA...`), emails, and phone numbers before data enters storage or LLM context.
- **Verbatim Grounding Check:** A candidate claim citing a fabricated Apollo GraphQL quote is caught by the grounding validator and dropped, logging a `GROUNDING_DROP` event.
- **Unanswered Question Uncertainty:** Dave's question regarding Twilio SMS gateway ownership received zero replies and is flagged as `UNCERTAIN / UNANSWERED` .
- **Orphaned Jira Ticket:** `HYP-107` (invoice reconciliation automation) has `Assignee: Unassigned` and is tagged `MISSING OWNER` .
- **Excluded Privacy Boundaries:** 3 sensitive streams (`#hr-confidential` , `dm-dave-sarah` , `hr-payroll-tools`) are automatically excluded at ingestion with explicit justifications shown in the UI.

04 Automated Test Evidence (Passing Test Suite)

The complete codebase is validated by 6 automated test suites executing against simulated database states (`pytest tests -v`):

TEST SUITE	VERIFICATION SCOPE	RESULT
`test_redaction.py`	Verifies pre-storage scrubbing of AWS keys, emails, phone numbers, and source exclusion rules.	**PASSED**
`test_grounding.py`	Confirms verbatim quote validation and dropping of ungrounded candidate claims.	**PASSED**
`test_supersession.py`	Validates Redis-to-Memcached temporal override, unanswered question uncertainty, and unassigned tickets.	**PASSED**
`test_roles_and_gating.py`	Enforces server-level `HTTP 403 Forbidden` for successors before `v1.0`, leaver-to-lead gating, and receipt.	**PASSED**
`test_audit.py`	Tests sequential SHA-256 hash chaining, detection of simulated block tampering, and clean restoration.	**PASSED**

05 Three Major Improvements Committed for Round 3

15 Points

In accordance with the Round 2 brief requiring *"substantial product improvements rather than cosmetic changes,"* Team InnoviX commits to delivering:

1. Semantic + LLM Contradiction Engine

Expand beyond predefined topics by combining embedding-based semantic claim clustering with an LLM evaluation pass that cross-examines arbitrary chat assertions against pull request diffs and commit histories across unconstrained project topics.

2. Role-Tailored Views with Sensitivity Levels

Implement granular claim sensitivity classifications (e.g. internal technical debt and debugging quirks visible strictly to technical successors, while executive roadmaps, risk registers, and ownership milestones surface in management views).

3. Interactive Day-1 Navigator with Audit Logs

Upgrade the successor view into a step-by-step onboarding walkthrough featuring direct evidence links, task progress tracking (checkboxes and verification checks), with completion milestones formally recorded into the cryptographic audit trail.

06 Known Limitations, Usability & Participant Details

10 Points Usability

AREA	FACTUAL DETAIL & ENGINEERING CONTEXT
Demo Role Picker	Roles are selected from a dropdown menu without verified logins to enable immediate evaluator inspection. Access controls are enforced by backend headers.
Extractor Scope	The deterministic rule-based extractor is calibrated to the planted Project Hyperion topics. Unconstrained topic extraction across arbitrary codebases relies on the Gemini LLM pipeline.
Data Corpus	Operates on simulated Project Hyperion fixtures (commits, PRs, chat, Jira) to maintain data privacy rather than live OAuth webhooks into corporate tools.
LLM Dependency	The Gemini extraction pipeline requires a valid `GEMINI_API_KEY` in the environment; if omitted or unavailable, the system defaults to deterministic extraction.
Free-Host Cold Starts	On free hosting services (e.g. Render free tier), inactive instances spin down and may experience a 30–50 second delay on initial load. A keep-alive script can mitigate this.
Ephemeral Database	On container platforms without attached persistent volumes, the SQLite database resides in container memory/ephemeral disk and resets if the container instance restarts.
AI Tooling Disclosure	In accordance with Rule 7, development used Google Antigravity / Gemini for assistance with code boilerplate, architecture planning, and synthetic dataset curation.
Participant & Team	**Team Name:** InnoviX | **Solo Builder & Lead:** Kavin S S G (Full-Stack AI Engineer).